

AI Acceptance Across Service Platforms

Data, Methodology and Progress against the Brief — Working Report

Htet Phone Khaing · Tatum Miranda Smith · Artem Cherkaskyy

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1 The idea

Online services increasingly put **AI and AI agents** between the user and the thing they came for: chatbots answer support tickets, algorithms screen and price, automated systems make decisions. We study **how people accept (or reject) that AI** — and, crucially, how acceptance changes with the *role* AI plays. We compare three service contexts:

- **AI-native services** — the product *is* an AI agent (ChatGPT, Claude, Replika, Gemini, Perplexity);
- **Peer-to-peer rentals** — AI is a *peripheral* automation layer on a human marketplace (Turo, Airbnb);
- **Customer service** — AI runs the *support* layer (chatbots replacing agents).

Research question. How do users accept and experience AI agents across different service platforms — and what drives or erodes trust as AI moves from being the product, to a marketplace add-on, to the support desk?

We use two complementary corpora. **Reddit is the discussion baseline** — open deliberation where users describe AI experiences in their own words. **Trustpilot is the verification layer** — reviews carrying a **1–5 star rating** that acts as a ground-truth acceptance label. Where the open discussion and the labelled reviews agree we gain confidence; where they diverge we learn something real.

2 Data source 1 — Reddit (discussion baseline)

The collection problem and fix. Reddit’s 2026 anti-bot wall blocks every anonymous data endpoint (HTTP 403), which defeats the usual R/RedditExtractor and .json-based scrapers alike. We therefore pull from the **Arctic Shift** research archive (a Pushshift successor) over its public API — archived Reddit data, no live blocking, no token or proxy (01_fetch_reddit_arcticshift.R). AI-native subreddits are pulled in full; rental and customer-service subreddits are pulled by AI/automation **keyword search** so we keep only on-topic comments. Every comment is tagged with a **platform_category**.

The dataset (reddit_baseline.csv): 4,269 comments across 12 subreddits, Nov 2023 – May 2026, mean length 374 characters. The top subreddits are r/turo (747), r/AirBnB (578), r/CustomerService (527), r/airbnb_hosts (390), r/artificial (270), r/replika (270). Comments split across three service contexts:

Table 1: Reddit comments by service context.

Platform category	Comments	Share
ai_service	2027	47%
rental	1715	40%
customer_service	527	12%

A relevance check confirms the corpus is genuinely on-topic (the previous, sharing-economy-targeted scrape was under 2%): customer-service comments are ~95% AI/automation-related and rental ~85% (keyword-filtered), while AI-native comments are on-topic by subreddit context (people in r/ClaudeAI or r/replika are discussing the AI product even when a comment omits the word “AI”).

3 Data source 2 — Trustpilot (verification)

Trustpilot reviews carry a **1–5 star rating on every review** — a ground-truth acceptance label we did not have to infer. We scraped seven peer-to-peer rental platforms via the Apify actor `casper11515/trustpilot-reviews-scraper` (web UI, once per company), merged them (`02_merge_trustpilot.R`), then flagged AI/automation reviews and categorised platforms (`06_ai_acceptance.R`).

The dataset: 3,912 reviews spanning Jul 2019 – May 2026. Of these, **60 describe an AI / automation experience** (bots, algorithms, automated decisions, “no human”). Those AI reviews average **2.32 stars vs 4.21** for the rest — on rental platforms, hitting automation correlates sharply with frustration. This is the project’s first concrete finding.

Extending the verification layer. The rental reviews give us AI-as-add-on. To match the Reddit AI-native context we are scraping **AI-native platforms** (`character.ai`, `replika.com`, `openai.com`, `perplexity.ai`) into the same pipeline; `06_ai_acceptance.R` auto-tags them `ai_service`, giving Trustpilot the same three-context structure as Reddit.

4 Classes and how the sources fit

For the brief’s per-class analysis our two **classes are positive vs negative acceptance** — on Trustpilot directly from the star rating (4–5 vs 1–2), on Reddit from lexicon sentiment validated against those stars. **platform_category** (AI-native / rental / customer-service) is the second lens applied to every figure. The triangulation: compute sentiment on Trustpilot text, validate it against the star label, apply the validated lexicon to Reddit, then compare which topics drive negative sentiment in each context. Agreement across two structurally different corpora is stronger evidence than either alone.

5 Progress against the assignment brief

Each task is tagged **Done**, **Partly done**, **In progress** or **To do**.

Dataset — Done. Two self-scraped corpora, no X/Twitter: Reddit (`reddit_baseline.csv`, 4,250 comments, 3 categories) and Trustpilot (3,912 star-labelled reviews + AI flag).

Cleaning — In progress. Collection strips deleted/removed, short and duplicate comments. Full normalisation (lowercase, punctuation/number/stop-word removal, tokenisation, lemmatisation, 1- and 2-grams) is the next script (`07_preprocess.R`).

Word frequency & clouds — **To do.** Top-10 words per class as bar charts; comparison and commonality clouds (uni- and bi-gram), per class.

Word networks — **To do.** Per-class co-occurrence networks (`widyr::pairwise_count` + `ggraph`).

Sentiment — **To do.** AFINN/NRC on both corpora, both classes on one graph, validated against Trustpilot stars (the convergent-validity check).

Topic modelling (LDA) — **To do.** 3–5 topics per class, 1- and 2-gram, with trust seed words and a `text2vec` cross-check.

Embeddings — **To do.** GloVe / Word2Vec neighbours of *trust*, *AI*, *agent*, *host* inside the topics.

Synthesis & literature — **To do / partly identified.** Map findings to trust and adoption frameworks (Mayer et al. 1995; Davis TAM; UTAUT), ending in data-backed **AI-agent design cards**. Owners to be assigned by the team.

6 Limitations

- Trustpilot skews positive (opt-in bias) — we down-sample for classification.
- The rental keyword pull includes a broad **app** query that adds some general app-experience noise; every Reddit row keeps a `source_query` column so it can be filtered precisely.
- AI-native Trustpilot platforms are still being scraped, so the Trustpilot `ai_service` context is not yet populated.
- “Trust” is mostly *implicit* (bot/human, automated, algorithm language), measured via topics and sentiment, not keyword counts. English-only.
- Reddit data is archival (Arctic Shift); the live anti-bot wall prevents fresh anonymous scraping, and Turo/Getaround per-platform review dates need a parse check before any per-platform time-series.

Repository: <https://github.com/tyomachkaa/ai-acceptance-sharing-economy> · Online Content Analysis, WU Vienna, 2026.